Type MS · Match digestive enzymes to substrate, product and site

Answer 1 of 12 · Question 2 part a.i · 3 marks

2023 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

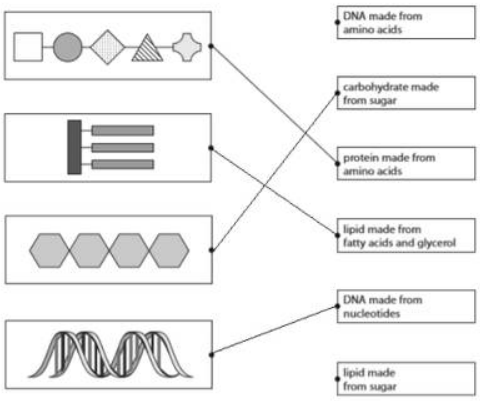
| | | | | |
|---|---|---|---|---|
| 2 (a) | (i) | in the following order:<br>• proteases;<br>• protein Q;<br>• amino acids; | Reject protein P | 3 |
| | (ii) | stomach/ small intestine/ pancreas/ duodenum/ ileum; | | 1 |
| | (iii) | DNA / mRNA / messenger RNA; | | 1 |
| (b) | (i) | • as pH increases activity increases;<br>• to pH2/peaks at pH2/to optimum;<br>• decreases from pH2; | Allow increases and then decreases for one mark | 3 exp |
| | (ii) | Any two from:<br>• enzyme is denatured (at pH2);<br>• change in shape of active sites;<br>• no enzyme-substrate complexes formed/no binding with substrate; | | 2 exp |
| | (iii) | • repeat;<br>• calculate a mean/average/remove anomalous result | | 2 grad |
| | (iv) | Any one from:<br>• temperature;<br>• substrate concentration/volume;<br>• enzyme concentration/volume; | | 1 grad |

**Total for Question 2 = 13 Marks**

Type MS · Match digestive enzymes to substrate, product and site

Answer 2 of 12 · Question 3 part b · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 | (a) |  | Do not allow more than one line from each structure | 3 |
| | (b) | **D** stomach; | | 1 |
| | (c) (i) | 3/three | | 1 |
| | (ii) | not washing equipment/transfer of food containing protein into distilled water; | Allow any valid alternative | 1 |
| | (d) | Broken glass - cuts: Clear up breakages using a dustpan and brush/keep equipment away from edge of tables; | Ignore wear gloves | 1 |
| | | Biuret reagent - irritant: Wear goggles/wash hands after use/don't use if skin is sensitive; | Ignore wear gloves | 1 |

**Total question 3 = 8 marks**

# Type MS · Match digestive enzymes to substrate, product and site

## Answer 2 of 12 · Question 3 part b · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Match digestive enzymes to substrate, product and site

### Answer 3 of 12 · Question 1 part b.i · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

| Question number | | | | |
|---|---|---|---|---|
| 1 | (a) | any three from<br>• oesophagus (1)<br>• stomach (1)<br>• small intestine/ileum/duodenum (1)<br>• large intestine/rectum (1) | | 3 |
| | (b) (i) | protease/pepsin/trypsin (1) | | 1 |
| | (ii) | • chewing/grinding/mastication (1)<br>• (mix with) saliva (1)<br>• digestion of starch (1)<br>• to maltose (1) | | 4 |
| | (c) | any three from<br>• provides bulk (1)<br>• helps peristalsis (1)<br>• prevents constipation (1)<br>• reduces risk of bowel cancer (1) | | 3 |

Total for question = 11 marks

## Type MS · Match digestive enzymes to substrate, product and site

### Answer 3 of 12 · Question 1 part b.i · 1 mark

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Match digestive enzymes to substrate, product and site

Answer 4 of 12 · Question 2 part a.i · 1 mark

2020 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) | C; (lipase) | | 1 |
| (ii) | A; (carbon, hydrogen, oxygen) | | 1 |
| (iii) | C; (liver) | | 1 |
| (b) | • show effect of enzyme (1)<br>• show that bile salts on their own don't break down fat (1) | Ignore control | 2 |
| (c) | • create alkaline/optimum pH (1)<br>• for enzyme to work quickly (1) | Allow increases/adjusts pH | 2 |
| (d) | • increase rate of digestion (1)<br>• with no bile salts change occurred after 15 minutes/digested more slowly in tube A (1)<br>• occurred after 5 minutes with bile salts present/digested more quickly in tube B (1) | | 3 |

Total 10 marks

## Type MS · Match digestive enzymes to substrate, product and site

### Answer 4 of 12 · Question 2 part a.i · 1 mark

2020 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

## Type MS · Match digestive enzymes to substrate, product and site

### Answer 5 of 12 · Question 2 part b.i · 1 mark

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 2 | (a) (i) | W large intestine;<br>X oesophagus;<br>Y stomach;<br>Z small intestine; | Allow colon<br><br>Allow ileum | 4 |
| | (ii) | In the following order only:<br>• bolus;<br>• muscles;<br>• contract;<br>• peristalsis; | | 4 |
| | (b) (i) | salivary gland/ pancreas; | | 1 |
| | (ii) | (lining of) **small** intestine/ ileum/ duodenum; | | 1 |
| | (iii) | • starch digested/broken down to sugar/glucose;<br>• (sugar) diffuses/moves out through (Visking tubing); | Allow maltose | 2 |
| | (iv) | Benedict's solution/reagent; | | 1 |
| | (v) | Any two from the following:<br>• blue/black colour;<br>• (undigested) starch is present;<br>• too large to pass through Visking tubing; | | 2 |

**Total marks 15**

## Type MS · Match digestive enzymes to substrate, product and site

### Answer 5 of 12 · Question 2 part b.i · 1 mark

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Match digestive enzymes to substrate, product and site

### Answer 6 of 12 · Question 4 part a.ii · 1 mark

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

| Question number | | | | |
|---|---|---|---|---|
| 4 | (a) (i) | Any three of the following:<br>• add alcohol/ethanol to food;<br>• shake;<br>• add water;<br>• milky coloration shows fat present; | Allow rub food;<br>onto filter paper;<br>translucent;<br>answer must be in sequence i.e. ethanol first | 3 |
| | (ii) | lipase | | 1 |
| | (iii) | fatty acids;<br>glycerol; | | 2 |
| | (b) (i) | Any value between 8.6 and 8.8; | | 1 |
| | (ii) | independent = time/ temperature;<br>dependent = pH; | | 2 |
| | (iii) | • 37°C is the optimum temperature;<br>• lipase/enzyme is more active/works faster;<br>• fat is digested more quickly;<br>• at 80 °C/higher temperature enzyme becomes denatured; | Allow reverse argument throughout:<br>• at 80°C/higher temperature enzyme is denatured;<br>• enzyme less active;<br>• fat digested more slowly;<br>• 37°C is the optimum temperature (for enzyme activity/enzyme works faster at 37°C | 4 |

**Total marks 13**

## Type MS · Match digestive enzymes to substrate, product and site

### Answer 6 of 12 · Question 4 part a.ii · 1 mark

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Match digestive enzymes to substrate, product and site

### Answer 7 of 12 · Question 4 part a.iii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 4 (a) (i) | Any three of the following:<br>• add alcohol/ethanol to food;<br>• shake;<br>• add water;<br>• milky coloration shows fat present; | Allow rub food;<br>onto filter paper;<br>translucent;<br>answer must be in sequence i.e. ethanol first | 3 |
| (ii) | lipase | | 1 |
| (iii) | fatty acids;<br>glycerol; | | 2 |
| (b) (i) | Any value between 8.6 and 8.8; | | 1 |
| (ii) | independent = time/ temperature;<br>dependent = pH; | | 2 |
| (iii) | • 37°C is the optimum temperature;<br>• lipase/enzyme is more active/works faster;<br>• fat is digested more quickly;<br>• at 80 °C/higher temperature enzyme becomes denatured; | Allow reverse argument throughout:<br>• at 80°C/higher temperature enzyme is denatured;<br>• enzyme less active;<br>• fat digested more slowly;<br>• 37°C is the optimum temperature (for enzyme activity/enzyme works faster at 37°C | 4 |

**Total marks 13**

## Type MS · Match digestive enzymes to substrate, product and site

### Answer 7 of 12 · Question 4 part a.iii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Match digestive enzymes to substrate, product and site

Answer 8 of 12 · Question 2 part a.ii · 1 mark

2023 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (a) | (i) | in the following order:<br>• proteases;<br>• protein Q;<br>• amino acids; | Reject protein P | 3 |
| | (ii) | stomach/ small intestine/ pancreas/ duodenum/ ileum; | | 1 |
| | (iii) | DNA / mRNA / messenger RNA; | | 1 |
| (b) | (i) | • as pH increases activity increases;<br>• to pH2/peaks at pH2/to optimum;<br>• decreases from pH2; | Allow increases and then decreases for one mark | 3 exp |
| | (ii) | Any two from:<br>• enzyme is denatured (at pH2);<br>• change in shape of active sites;<br>• no enzyme-substrate complexes formed/no binding with substrate; | | 2 exp |
| | (iii) | • repeat;<br>• calculate a mean/average/remove anomalous result | | 2 grad |
| | (iv) | Any one from:<br>• temperature;<br>• substrate concentration/volume;<br>• enzyme concentration/volume; | | 1 grad |

**Total for Question 2 = 13 Marks**

Type MS · Match digestive enzymes to substrate, product and site

Answer 9 of 12 · Question 4 part a.vi · 1 mark

2024 JUN · Paper 01 (regional R) · find part a.vi in this mark scheme extract

| Question number | | | | |
|---|---|---|---|---|
| 4 (a) | (i) | • axes labels and units and right way around;<br>• linear scale and at least half grid;<br>• correctly plotted points to half square tolerance; | | (3) |
| | (ii) | • curve of best fit; | do not accept straight line going through all points | (1) |
| | (iii) | 6 | | (1) |
| | (iv) | • time taken to digest starch decreases until pH 6;<br>• time taken to digest starch increases (from/after PH6) to pH10; | Allow 1 mark for decrease and then increase in time as pH increases | (2) |
| | (v) | • reference to pH2 not being optimum/best pH for **enzyme** (activity)/pH6 being (closer to) optimum pH for enzyme;<br>• enzymes denatured / active site changes shape/deformed (at pH2);<br>• less/no binding to substrate/ substrate cannot fit into active site/ fewer enzyme -substrate complex; | | (3) |
| | (vi) | small intestine/ mouth/duodenum; | | (1) |
| (b) | | • add iodine solution;<br>• blue-black/blue/black colour indicates the presence of starch; | | (2) |

**Total for Question 4 = 13 marks**

Type MS · Match digestive enzymes to substrate, product and site

Answer 10 of 12 · Question 1 part b · 4 marks

2024 JUN · Paper 02 (regional R) · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (a) (i) | | X = incisor(s);<br>Z = molar(s)/pre-molar(s); | | (2) |
| | (ii) | tearing/ripping/gripping/holding (food); | | (1) |
| (b) | | • amylase/carbohydrase;<br>• sugar/maltose;<br>• any value between 6.7 and 7.4;<br>• decrease/reduce/stop/slow down/take more time; | allow glucose | (4) |
| (c) (i) | | oesophagus; | | (1) |
| | (ii) | • muscular contractions/peristalsis;<br>• push food/bolus (downwards/towards stomach); | | (2) |

**Total for Question 1 = 10 marks**

Type MS · Match digestive enzymes to substrate, product and site

Answer 11 of 12 · Question 7 part f · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part f in this mark scheme extract

| | | | |
|---|---|---|---|
| 7 (a) | • body temperature;<br>• optimum/enzyme is not denatured; | Accept alimentary for MP1 | 2 |
| (b) | • effect of pH/to see if pH affects enzyme activity;<br>• range of pH used/different pH used/ pH is the independent variable; | ***Effect of different pH on enzyme activity = 2 marks*** | 2 |
| (c) | Any four from:<br><br>Answers about validity( MP1-4)<br>• The same temperature;<br>• The same volume of enzyme/suspension;<br>• not (valid) because different time used;<br>• the same enzyme is used;<br><br>Answers about accuracy( MP 5-7)<br>• as difficult to judge endpoints/ when it goes clear;<br>• students' judgement of endpoint may vary/ subjective/ not standardised/ did not use a colorimeter;<br>• timing is too short to reach endpoint/ cannot tell it will go clear if left for longer; | They do not have to mention validity/accuracy to score marks.<br><br>Must have at least one mark from each section to score 4. Maximum of 3 if answers are from one section | 4 |
| (d) | • tube with no enzyme/ denatured enzyme/boiled enzyme;<br>• same volume of boiled enzyme/water /5cm$^3$ of water/boiled enzyme;<br>• same volume of suspension/same overall volume/ same volume of substrate ( in test tube); | | 3 |
| (e) | • optimum pH;<br>• fastest between 2 and 4; | | 2 |
| (f) | • stomach;<br>• acidic/low pH/pH2-4/pH2; | | 2 |

**Total 15 marks**

## Type MS · Match digestive enzymes to substrate, product and site

### Answer 11 of 12 · Question 7 part f · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part f in this mark scheme extract

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## Type MS · Match digestive enzymes to substrate, product and site

### Answer 12 of 12 · Question 7 part a · 1 mark

2025 JUN · Paper 02 (regional R) · find part a in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 7 | (a) | amylase; | | 1 |
| | (b) | Any two from<br>• enzymes are immobilised on testing strip;<br>• react with/breaks down glucose (in urine);<br>• (catalyse a chemical reaction that) causes a colour change (in strip); | | 2 |
| | (c) | pancreas; | | 1 |
| | (d) | • rejection/immune response/allergies;<br>• (immune system) recognised it as foreign/antigen; | | 2 |
| | (e) | any two from<br>• use restriction enzymes;<br>• cut out insulin gene from human DNA;<br>• insert human insulin gene into yeast genome/plasmid/genetic material; | | 2 |
| | | | **Total for Question 7 = 8 marks** | |